Knowledge Base Entry — Human-Centered Data Platform Engineering

Configuration Specialist (Contract) — H. Lundbeck A/S

Role Overview

Embedded consultant on Lundbeck's **Research Data Platform (RDP)**, contributing as a fully integrated member of a cross-functional project team. The role is organized around **use-case–driven project work**, spanning discovery, solution design, implementation, verification, and go-live. Although formally titled "Configuration Specialist," the role combines **data platform engineering, business analysis, and user advocacy**, with a strong emphasis on scientific context, data integrity, and real-world usability.

Project-Based Working Model

- Worked in a **use-case–oriented delivery cycle**, typically handling one primary use case at a time (occasionally overlapping), covering:
    - Discovery sessions with stakeholders and end users
    - Solution design and requirements clarification
    - Implementation and configuration of data pipelines
    - Testing, verification, and preparation for go-live
    - Hypercare and post-deployment support
- Maintained continuity across phases, acting as one of the few roles with **end-to-end visibility** of scope, risks, progress, and dependencies.
- Often served as an informal knowledge hub due to being consistently present across technical, business, and user-facing discussions.

Data Pipelines, Configuration & Technical Contributions

- Developed and configured automated pipelines for ingestion, annotation, transformation, and processing of laboratory instrument data into a **TetraScience-based Research Data Platform**.
- Worked extensively with **Intermediate Data Schemas (IDS)**:
    - Vendor-agnostic JSON schemas used to map raw, vendor-specific instrument data into standardized, validated, and hierarchically structured representations.
    - IDS defines naming, data types, allowed ranges, hierarchy, and semantic meaning to ensure predictability, consistency, and downstream usability.
- Contributed to the design and evolution of IDS definitions in collaboration with stakeholders, ensuring that standardized representations remained **faithful to the original experimental data**.
- Implemented and reviewed pipelines that generate IDS-based JSON outputs aligned with FAIR data principles.
- Reviewed and validated deliverables from external vendors to ensure:

    - Alignment with agreed solution designs and RFPs
    - Consistency with IDS definitions and semantic expectations
    - Sufficient data integrity for downstream querying, aggregation, and analysis
- Investigated technical limitations in validation strategies (e.g. partially unvalidated structured data formats), surfaced associated risks, and helped establish **explicit, documented agreements** on structure, content, and acceptable trust-based handling when no better technical alternative existed.

In practice, “configuration” meant defining and validating **vendor-agnostic data representations** that preserve the semantic meaning of raw instrument data while enabling reliable downstream use.

User-Centered & Business-Focused Work

- Considered **laboratory scientists and instrument operators** as the primary users of the platform.
- Engaged directly with users to understand:
    - Real laboratory workflows
    - Operational constraints
    - The difference between “how things are usually done” and “what is actually possible”
- Identified and surfaced gaps between user expectations, system behavior, and data reality, including cases where:
    - Instrument outputs lacked essential protocol metadata
    - Automated ingestion still resulted in **non-reproducible experiments**
- Acted as a bridge between:
    - End users and vendors
    - Business expectations and technical feasibility
- Provided practical, operational feedback on discovery and solution design documentation, identifying missing assumptions, edge cases, and downstream consequences before they became production issues.

Documentation, Enablement & Go-Live Support

- Created **user guides and workflow documentation** to support adoption of new functionality entering production.
- Supported go-live activities and hypercare periods, ensuring users could navigate new workflows with confidence and minimal manual overhead.
- Treated documentation as a **core delivery artifact and risk-mitigation mechanism**, not as an afterthought.

Professional Judgment & Way of Working

- Regularly balanced ideal technical solutions against real-world constraints, raising concerns clearly while accepting informed trade-offs when required.
- Demonstrated strong awareness that:
    - Manual steps increase sensitivity to error
    - Poorly structured data undermines long-term scientific value
    - Knowledge silos and undocumented assumptions create operational risk
- Preferred holistic problem-solving over isolated implementation, combining technical depth with empathy for users and respect for scientific context.
- Worked most effectively in environments where insight-sharing, transparency, and cross-disciplinary dialogue were encouraged and valued.

Reusable Professional Insights

- “I work best when I can combine technical implementation with an understanding of how systems are actually used in practice.”
- “Making data FAIR is not only a technical challenge—it requires understanding experiments, instruments, and the people who operate them.”
- “Standardizing data is only valuable if the standardized representation remains faithful to the original experiment; otherwise, automation can silently erode scientific trust.”

Core Keyword Cluster

Research Data Platforms · FAIR Data · Intermediate Data Schemas (IDS) · Scientific Data Integrity · Instrument Data · Semantic Data Modeling · User-Centered Engineering · Business Analysis · Vendor Collaboration · Regulated Environments · Documentation & Enablement